

MAPLE LAKE, MN, USA

WMO: 722144

Lat: 45.236N

Lon: 93.986W

Elev: 313

StdP: 97.62

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04945

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.0	(c) -22.5	(d) -28.5	(e) 0.3	(f) -24.4	(g) -26.3	(h) 0.4	(i) -22.2	(j) 10.1	(k) -10.1	(l) 8.8	(m) -7.9	(n) 2.1	(o) 300	(p) 0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	31.2	23.2	29.8	22.5	28.2	21.3	25.0	29.2	23.9	28.1	22.8	26.7	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.3	28.0	22.5	17.8	26.7	21.4	16.7	25.4	78.0	29.3	73.2	28.2	68.8	26.7	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.8	(b) 7.9	(c) 7.0	(e) -29.4	(f) 34.3	(g) 2.9	(h) 1.6	(i) -31.5	(j) 35.5	(k) -33.2	(l) 36.4	(m) -34.9	(n) 37.3	(o) -37.0	(p) 38.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-9.6	-8.4	-0.3	7.2	14.2	19.7	22.1	20.4	16.5	8.6	1.0	-7.0		
	DBStd	12.44	7.33	6.62	6.87	5.50	4.94	3.43	2.98	3.08	4.54	4.89	5.78	6.50		
	HDD10.0	2470	609	516	324	115	16	0	0	0	4	86	273	528		
	HDD18.3	4414	867	749	577	334	147	24	5	14	87	303	520	786		
	CDD10.0	1418	0	0	6	32	146	292	375	321	200	43	3	0		
	CDD18.3	319	0	0	0	1	19	66	122	77	33	2	0	0		
	CDH23.3	2683	0	0	1	13	214	534	1058	591	249	23	0	0		
	CDH26.7	716	0	0	0	1	66	142	312	137	56	3	0	0		
(14) Wind	WSAvg	2.8	3.0	3.1	3.1	3.5	3.0	2.4	2.0	1.9	2.3	2.9	3.0	2.9		
(15) Precipitation	PrecAvg	797	19	18	39	72	93	121	98	108	92	65	41	23		
	PrecMax	1292	50	42	91	202	196	338	263	217	211	168	120	57		
	PrecMin	545	2	3	8	16	21	48	25	17	15	15	2	6		
	PrecStd	158	14	11	21	43	43	63	57	54	56	41	31	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	10.9	20.4	25.2	32.4	32.6	33.5	31.7	30.8	26.3	18.3	8.3		
		MCWB	3.1	9.0	14.1	15.0	20.2	22.9	25.6	24.2	23.2	17.6	12.6	6.4		
	2%	DB	3.4	5.4	15.8	21.4	28.1	30.1	31.3	29.8	28.3	22.6	14.8	4.8		
		MCWB	1.8	3.1	11.5	12.8	18.4	21.8	23.8	23.4	21.4	15.7	10.1	2.9		
	5%	DB	1.8	2.8	12.3	19.0	25.6	28.0	29.9	28.0	26.4	19.5	11.6	2.7		
		MCWB	0.9	1.3	9.3	11.9	17.3	20.5	23.0	21.6	20.4	14.2	8.2	1.7		
	10%	DB	0.2	0.8	8.8	16.5	22.8	26.4	28.2	26.8	24.2	16.5	9.0	1.3		
		MCWB	-0.5	-0.3	6.2	10.2	15.9	19.4	22.0	20.6	19.4	12.3	6.5	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	8.8	16.2	16.1	22.8	25.2	27.1	25.7	24.9	20.3	14.0	7.0		
		MCDB	5.5	11.2	19.4	22.7	29.6	30.0	30.8	30.1	28.7	22.9	17.6	7.9		
	2%	WB	2.0	3.5	12.3	14.3	20.3	23.4	25.4	24.5	22.9	16.8	10.9	3.1		
		MCDB	3.2	5.2	15.1	19.3	25.0	27.9	29.6	28.4	26.5	20.9	14.3	4.3		
	5%	WB	0.8	1.4	9.1	12.7	18.7	21.9	24.2	23.2	21.5	14.9	8.4	1.8		
		MCDB	1.6	2.7	12.5	17.7	23.5	26.5	28.6	26.7	24.7	18.6	11.1	2.6		
	10%	WB	-0.4	-0.1	6.2	10.9	17.1	20.6	23.0	22.0	20.1	12.8	6.6	0.6		
		MCDB	0.1	0.8	8.6	15.6	21.7	24.8	27.2	25.0	23.7	15.9	9.0	1.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.8	9.5	10.6	11.2	10.4	10.8	10.6	10.9	9.6	8.3	7.6		
		MCDBR	9.4	10.7	13.2	15.2	15.1	12.5	11.8	11.7	12.5	14.3	12.3	8.1		
	5% WB	MCWBR	7.9	8.7	8.6	8.6	7.3	6.1	5.9	6.0	6.7	8.3	8.1	6.9		
		MCWBR	9.2	10.4	12.2	13.2	12.6	10.7	10.6	10.2	11.0	11.9	11.3	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.276	0.311	0.367	0.401	0.392	0.389	0.400	0.368	0.315	0.286	0.261		
		taud	2.415	2.373	2.395	2.311	2.271	2.342	2.370	2.334	2.392	2.536	2.514	2.444		
	Ebn at Noon		862	915	924	892	868	876	874	850	850	862	832	830		
		Edh at Noon	71	89	102	121	131	122	118	118	102	77	65	62		
(44) All-Sky Solar Radiation	RadAvg		1.73	2.81	3.81	4.65	5.36	5.96	6.35	5.29	4.09	2.60	1.66	1.22		
	RadStd		0.21	0.30	0.35	0.43	0.43	0.44	0.22	0.41	0.29	0.35	0.15	0.15		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page